A Hamiltonian Dispersion Framework for Kerr Photon Rings, Frequency-Dependent Shadow Constraints, Superradiance, and Eikonal Quasinormal Modes

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Abstract

We develop a Hamiltonian dispersion framework for massless wave propagation in Kerr spacetime with a quartic local-frame correction,

$$H = \tfrac{1}{2}\big[g^{\mu\nu}k_\mu k_\nu + \Lambda\,(k_{\rm loc}^{\rm ZAMO})^4\big],$$

and derive four interconnected results. **(1) Photon-ring correction:** the impact parameter shifts as $\delta b = C(a^*, \text{ring})\,\Lambda E^2$ with $C_{\rm pro} = -0.6230\,M^{-1}$ and $C_{\rm ret} = +11.495\,M^{-1}$ for $a^* = 0.9$; the E^2 scaling is proved analytically, and the large $|C_{\rm ret}/C_{\rm pro}| \approx 18.5$ asymmetry is traced to near-degeneracy of the ring-defining Jacobian at the retrograde ring (Section 3), not to the local dispersive momentum $k_{\rm loc}^4$ itself. **(2) Chromatic shadow:** the shadow size is frequency-dependent, with $\delta\theta \propto \Lambda\,\omega^2/D$, and the frequency-dependent sensitivity scale satisfies $\Lambda_{\max} \propto \omega^{-2}$ (exact algebraic scaling, confirmed numerically to slope -2.0000). **(3) Superradiance:** we prove that the kinematic threshold $\omega = m\Omega_H$ is not modified by the quartic dispersion; the amplification factor away from threshold is not addressed here. **(4) Eikonal QNMs:** we use an eikonal continuation of the classical photon-orbit/QNM correspondence, $\omega_{\rm QNM} \simeq \Omega_{\rm ph}(\Lambda)\,l - i(n+\tfrac{1}{2})\lambda_{\rm Lyap}(\Lambda)$, as an effective approximation, with dispersive corrections $\delta\Omega_{\rm ph}/\Omega_{\rm ph}^{\rm GR} \simeq +0.657\,\Lambda E^2$ (prograde) and $+5.05\,\Lambda E^2$ (retrograde) for $a^* = 0.9$; since the photon energy of an eikonal mode is its frequency, $E \simeq l\,\Omega_{\rm ph}$, these corrections grow as l^3. The $\Lambda \to 0$ limit is validated against exact Schwarzschild and Kerr benchmarks. These results provide a self-consistent theoretical framework for deriving observational tests and frequency-dependent sensitivity estimates for the dispersion parameter Λ through black-hole shadow chromaticity, ring asymmetry, and gravitational-wave ringdown, without yet constituting a measured or fitted observational bound.

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1 Introduction

In a companion paper [Kretski, 2026a] (hereafter Paper 1), we introduced a one-parameter Hamiltonian model for dispersive massless-wave propagation in Schwarzschild spacetime and proved two theorems on the non-equivalence of field-theoretic and worldline descriptions under quartic local-frame corrections. The present paper extends the framework to Kerr spacetime and derives four observational results.

The central Hamiltonian is

$$H = \tfrac{1}{2}\big[g^{\mu\nu}k_\mu k_\nu + \Lambda\,(k_{\rm loc}^{\rm ZAMO})^4\big], \tag{1}$$

where $k_{\rm loc}^{\rm ZAMO}$ is the locally measured wavenumber with respect to the ZAMO observer (Section 2), and $\Lambda > 0$ has dimensions of length squared.

We address four questions:

(Q1) How does Λ shift the photon-ring impact parameter, and what is the energy dependence?

(Q2) Is the black-hole shadow chromatic?

(Q3) Does the superradiant threshold $\omega < m\Omega_H$ receive a Λ-dependent correction?

(Q4) What is the eikonal quasinormal-mode (QNM) prediction?

Summary of results. Q1 and Q2 are answered by a single analytical result: $\delta b = C\,\Lambda\,E^2$ (Section 3 and 4). Q3 is answered negatively by an exact proof (Section 6). Q4 is answered in the eikonal ($l \gg 1$) limit (Section 7).

The Event Horizon Telescope has produced resolved shadow images of M87* [EHT Collaboration, 2019] and Sgr A*, providing the observational context for photon-ring-based tests of strong-field gravity. Lorentz-invariance- violating dispersion relations have been constrained via Fermi-LAT gamma-ray-burst timing [Abdo et al., 2009] and are classified within the Standard Model Extension framework [Liberati, 2013]. The eikonal correspondence between photon-ring orbital/Lyapunov parameters and black-hole quasinormal modes was established by Cardoso et al. [2009] and reviewed by Berti et al. [2009]; we adopt this correspondence as the basis for Section 7.

2 Setup: Kerr + Quartic Dispersion

2.1 Kerr metric and ZAMO observer

We use Boyer-Lindquist coordinates with signature $(-,+,+,+)$, $c = 1$, $G = 1$, and natural units $M = 1$ (results are easily dimensionalised). The spin parameter satisfies $a^* = a/M \le 1$.

The ZAMO (zero-angular-momentum observer) has zero angular momentum, $g_{t\phi}n^t + g_{\phi\phi}n^\phi = 0$, giving angular velocity

$$\Omega_{\rm ZAMO} = \frac{n^\phi}{n^t} = -\frac{g_{t\phi}}{g_{\phi\phi}}, \tag{2}$$

and 4-velocity

$$n^\mu = \big(n^t,\,0,\,0,\,\Omega_{\rm ZAMO}\,n^t\big), \qquad n^t = \big(-g_{tt} - 2g_{t\phi}\Omega_{\rm ZAMO} - g_{\phi\phi}\Omega_{\rm ZAMO}^2\big)^{-1/2}, \tag{3}$$

normalised so that $g_{\mu\nu}n^\mu n^\nu = -1$. This construction remains valid in the ergosphere, provided r is outside the event horizon — unlike the static observer, which ceases to exist there. Contracting with the wave covector $k_\mu = (-\omega,\, k_r,\, k_\theta,\, m)$ gives

$$n^\mu k_\mu = n^t \left(-\omega + m\,\Omega_{\rm ZAMO}\right). \tag{4}$$

This is essential because the prograde photon ring of a rapidly spinning Kerr black hole lies inside the ergosphere (for $a^* = 0.9$: $r_{\rm ph}^{\rm pro} = 1.558\,M < r_{\rm erg}^{\rm eq} = 2M$).

2.2 Local wavenumber and Hamiltonian

The locally measured wavenumber squared is

$$(k_{\rm loc}^{\rm ZAMO})^2 = g^{\mu\nu}k_\mu k_\nu + (n^\mu k_\mu)^2, \tag{5}$$

which reduces correctly to k^2/α^2 in the Schwarzschild equatorial limit (Paper 1, Appendix A). The Hamiltonian (1) satisfies $H = 0$ (null constraint), and reduces to the standard Kerr null geodesic at $\Lambda = 0$.

3 Photon Ring: First-Order Correction

3.1 GR benchmark

For $\Lambda = 0$, the circular photon orbits satisfy $H = 0$ and $\partial H/\partial r = 0$. In the equatorial plane of Kerr with $a^* = 0.9$, this gives [Bardeen, 1973]:

$$r_{\rm ph}^{\rm pro} = 1.557855\,M, \qquad b_{\rm ph}^{\rm pro} = 2.844421\,M, \tag{6}$$

$$r_{\rm ph}^{\rm ret} = 3.910268\,M, \qquad b_{\rm ph}^{\rm ret} = -6.832319\,M. \tag{7}$$

These match the Bardeen polynomial $r^3 - 3Mr^2 \pm 2a\sqrt{Mr^3} = 0$ to residual $< 3 \times 10^{-7}$. This reproduction of the $\Lambda \to 0$ limit constitutes the GR validation of the numerical implementation used throughout this section.

3.2 Dispersive correction

Result 1 (Photon-ring shift)**.** *For $\Lambda > 0$, the impact parameter satisfies*

$$b_{\rm ph}(E, \Lambda) = b_{\rm GR} + C(a^*,\, \text{ring})\,\Lambda\,E^2 + O(\Lambda^2), \tag{8}$$

where E is the photon energy in units $Mc^2/\hbar$, and

$$C_{\rm pro} = -0.6230\ M^{-1}, \qquad C_{\rm ret} = +11.495\ M^{-1}. \tag{9}$$

The ratio $|C_{\rm ret}/C_{\rm pro}| = 18.45$ is universal for $a^* = 0.9$.

The coefficient C is obtained from implicit differentiation of the system $\{H = 0,\, \partial H/\partial r = 0\}$ with respect to Λ, giving

$$C = -\frac{J_{21}\,\text{rhs}_1 - J_{11}\,\text{rhs}_2}{\det J}, \tag{10}$$

where J is the Jacobian of the system and $\text{rhs}_1 = -\frac{1}{2}(k_{\rm loc}^2)^2$ at the GR ring. The signs are $C_{\rm pro} < 0$ (shadow shrinks) and $C_{\rm ret} > 0$ (shadow grows), reflecting the opposite orientations of the two orbit families relative to the frame-dragging.

Remark 1 (Origin of the $C_{\rm pro}/C_{\rm ret}$ magnitude asymmetry)**.** *The large asymmetry between $C_{\rm pro}$ and $C_{\rm ret}$ (a factor ≈ 18.5) is* not *primarily a consequence of the local dispersive momentum $k^4_{\rm loc}$, which is in fact larger at the retrograde ring by only a factor of ≈ 2.2 ($k^2_{\rm loc} = 2.827$ retrograde vs. 1.892 prograde) and would, taken alone, predict a* smaller *shift there. Since $\partial_r H = 0$ identically at any photon ring (the defining condition $J_{11} \equiv \partial_r H = 0$), the determinant in Eq. (10) reduces to $\det J \approx -(\partial_b H)(\partial_r^2 H)$. Direct numerical evaluation gives $|\det J|$ smaller at the retrograde ring by a factor of ≈ 565 relative to prograde, driven mainly by $\partial_r^2 H$ being ≈ 68 times smaller there — i.e., a flatter radial effective potential at the retrograde ring. This near-degeneracy of the ring-defining Jacobian, rather than the magnitude of $k^4_{\rm loc}$, dominates the observed asymmetry. The flatness of the radial potential does not by itself set the orbital Lyapunov exponent of Section 7, which also depends on $\partial^2 H/\partial k_r^2$ and on $\dot{t}$: in coordinate time the retrograde ring has the* larger *exponent ($\lambda_{\rm Lyap} = 0.1865$ versus 0.1361 at $\Lambda = 0$). (Version 0.4 predicted the opposite; see Section 8.)*

Table 1 gives numerical values for selected Λ.

Table 1: Impact parameter $b_{\rm ph}$ vs Λ for $a^* = 0.9$, $M = 1$, $E = 1$.

Λ	$r_{\rm pro}$	$b_{\rm pro}$	$r_{\rm ret}$	$b_{\rm ret}$
0	1.557855	+2.844421	3.910268	−6.832319
10^{-4}	1.557880	+2.844359	3.909626	−6.831170
10^{-3}	1.558104	+2.843799	3.903876	−6.820868
10^{-2}	1.560339	+2.838235	3.849229	−6.721647

4 Chromatic Shadow

4.1 Analytical proof of E^2 scaling

Theorem 1 (Chromatic shadow)**.** *The first-order correction $\partial b/\partial\Lambda|_{\Lambda=0}$ is exactly proportional to E^2:*

$$\left.\frac{\partial b_{\rm ph}}{\partial \Lambda}\right|_{\Lambda=0} = C\,E^2. \tag{11}$$

Proof. At the GR photon ring, $H_{\rm GR} = 0$ implies $g^{\mu\nu}k_\mu k_\nu = 0$. Therefore $k^2_{\rm loc} = (n^\mu k_\mu)^2 = E^2\,K^2$, where $K = -n^t + n^\phi b$ is an E-independent geometric factor. Consequently $k^4_{\rm loc} = E^4K^4$, all Jacobian entries scale as E^2, and the implicit-differentiation formula gives $\partial b/\partial\Lambda = C(\text{geom}) \cdot E^2$ exactly. □

4.2 Numerical confirmation

The power-law fit over five decades of E gives exponent $n = 2.000$ for both prograde and retrograde channels. The GR control (at $\Lambda = 0$) is achromatic to $\sigma(b_{\rm GR}) < 10^{-14}$ across $E \in [0.3, 10]$.

4.3 Physical formula (SI)

To restore physical units unambiguously, define the dimensionless dispersion parameter and photon energy

$$\hat{\Lambda} \equiv \frac{\Lambda_{\rm SI}}{M_{\rm geom}^2}, \qquad M_{\rm geom} \equiv \frac{GM}{c^2}, \qquad \mathcal{E} \equiv \frac{\hbar\omega}{Mc^2}, \tag{12}$$

so that all coefficients $C_{\rm pro}, C_{\rm ret}$ quoted above are the dimensionless C in $\delta b/M_{\rm geom} = C\,\hat{\Lambda}\,\mathcal{E}^2$. Reinstating physical units:

$$\delta\theta_{\rm shadow} = C\,\frac{GM}{c^2}\,\frac{\Lambda_{\rm SI}\,(\hbar\omega)^2}{M^2c^4\,D}, \tag{13}$$

where D is the distance to the black hole. This formula enables a direct test: measuring the shadow size at two radio frequencies ω_1, ω_2 gives

$$\Lambda_{\rm SI} = \frac{\delta b(\omega_2) - \delta b(\omega_1)}{C\,(GM/c^2)\,(\hbar/Mc^2)^2\,(\omega_2^2 - \omega_1^2)}. \tag{14}$$

5 Frequency-Dependent Observational Sensitivity

From (13) and the adopted 10% shadow-size tolerance (calibrated to EHT M87* uncertainty at 230 GHz):

Fixed-tolerance scaling (A3.3) — primary theoretical result:

$$\Lambda_{\rm max}(\nu) = A\,\nu^{-2}, \tag{15}$$

with numerical values in Table 2. The measured slope -2.0000 reproduces the exact ω^{-2} algebraic scaling derived in Theorem 1.

Table 2: Illustrative fixed-tolerance sensitivity scale for Λ (retrograde channel). These values are *not* observational constraints; they represent the value of Λ corresponding to the adopted 10% angular-tolerance model, not a fit to actual EHT data or emission physics. See text.

Frequency	$\Lambda_{\rm max}$ [m^2]
86 GHz	1.88×10^{-8}
129 GHz	8.37×10^{-9}
230 GHz	2.63×10^{-9}
345 GHz	1.17×10^{-9}

Remark 2. *Table 2 is an internal consistency test of the model scaling, not an independent EHT observational bound. A rigorous constraint requires per-band sensitivity modelling, atmospheric opacity, and imaging systematics of the EHT array.*

Resolution-scaled toy extension (A3.4) — illustrative only: Adopting a diffraction-limited toy model for the per-band tolerance gives an effective slope of -2.495 (steeper than the algebraic -2). This result depends on an assumed diffraction-limited ring-fraction model and does not incorporate frequency-dependent instrumental sensitivity,

opacity, or calibration systematics of actual observations. A3.3 is the theoretical scaling result of this paper; A3.4 is retained solely as an illustrative sensitivity-modelling extension and should not be read as a physical prediction.

TeV feasibility (A3.3 companion test): A separate feasibility analysis confirms the theoretical E^2 chromatic-shift scaling numerically (measured slope -2.000000 against the expected -2.000000) while showing that high photon energy alone does not provide spatial resolution: a direct observational constraint on Λ requires the energy-dependent photon-ring displacement to be spatially resolved, which current TeV instruments do not provide model-independently. Model-dependent spectral or flux constraints are not excluded and would require a dedicated emission and propagation model, which is beyond the scope of this paper.

Comparison with Fermi LAT: From Paper 1, $\Lambda \lesssim \mathcal{O}(10^{-53})\,\mathrm{m}^2$ from quadratic LIV bounds [Abdo et al., 2009]. The EHT radio sensitivity scales in Table 2 are weaker by ~ 45 orders of magnitude, because radio photons have $E_{\mathrm{photon}} \ll M_{\mathrm{BH}}c^2$.

6 Superradiance

Theorem 2 (Superradiant threshold)**.** *The superradiant threshold $\omega = m\Omega_H$ is not modified by the quartic dispersion $\Lambda\, k_{\mathrm{loc}}^4$:*

$$\omega_{\mathrm{thresh}}(\Lambda) = m\Omega_H + O(\varepsilon_{\mathrm{num}}), \tag{16}$$

where $\varepsilon_{\mathrm{num}}$ denotes the numerical proximity $r = r_+ + \epsilon$.

Proof. At $r = r_+$, the ZAMO angular velocity satisfies $\Omega_{\mathrm{ZAMO}} \to \Omega_H$ exactly (since $\Delta \to 0$). Consequently, at $\omega = m\Omega_H$:

$$n^\mu k_\mu = -n^t\,\omega + n^\phi\, m = n^t\, m\,(-\Omega_H + \Omega_{\mathrm{ZAMO}}) \to 0. \tag{17}$$

Therefore $k_{\mathrm{loc}}^2 \to 0$, and $\Lambda\, k_{\mathrm{loc}}^4 \to 0$ identically at the threshold. The threshold condition $\omega = m\Omega_H$ is therefore unchanged. □

Remark 3. *This result holds for the kinematic threshold only. The rate of energy extraction (amplification factor) may receive Λ-dependent corrections, which require computing the full transmission amplitude of the dispersive wave equation.*

7 Eikonal Quasinormal Modes

7.1 Framework

We use an eikonal continuation of the classical photon-orbit/QNM correspondence as an effective approximation for the dispersive Hamiltonian. In the eikonal limit $l \gg 1$, the QNM frequency is estimated by [Mashhoon, 1985, Cardoso et al., 2009]:

$$\omega_{\mathrm{QNM}} \simeq \Omega_{\mathrm{ph}}(\Lambda)\, l - i\big(n + \tfrac{1}{2}\big)\lambda_{\mathrm{Lyap}}(\Lambda), \tag{18}$$

where the orbital frequency and the Lyapunov exponent follow from Hamilton's equations of the *full* dispersive Hamiltonian,

$$\Omega_{\mathrm{ph}} = \frac{\dot\phi}{\dot t} = \frac{\partial H/\partial k_\phi}{\partial H/\partial k_t}, \qquad \lambda_{\mathrm{Lyap}} = \frac{\sqrt{-\,\partial_{k_r}^2 H\;\partial_r^2 H}}{|\partial H/\partial k_t|}, \tag{19}$$

evaluated on the circular orbit ($k_r = 0$, $H = \partial_r H = 0$); the second expression follows from linearising $\dot{r} = \partial H/\partial k_r$, $\dot{k}_r = -\partial H/\partial r$ about the orbit. Both quantities depend on Λ through the shift of the ring *and* through the dispersive terms of H, and, like δb (Theorem 1), their first-order corrections are proportional to ΛE^2. In the eikonal correspondence the photon energy is the mode frequency, $E \simeq \mathrm{Re}\,\omega_{\mathrm{QNM}} \simeq l\,\Omega_{\mathrm{ph}}$. A first-principles derivation of the dispersive QNM spectrum from the underlying radial wave equation — required to assess the validity of this continuation beyond the strict eikonal regime — is deferred to Paper 3.

7.2 GR validation

The $\Lambda \to 0$ limit of the framework is validated against three exact benchmarks:

Test 1 (Schwarzschild): $r_{\mathrm{ph}} = 3M$, $b_{\mathrm{ph}} = 3\sqrt{3}\,M$, $\Omega_{\mathrm{ph}} = \lambda_{\mathrm{Lyap}} = 1/(3\sqrt{3}\,M)$, reproduced to ten significant digits with Eq. (19). ✓

Test 2 (Kerr, $a^* = 0.9$): $r_{\mathrm{ph}}^{\mathrm{pro}}$, $b_{\mathrm{ph}}^{\mathrm{pro}}$ match Bardeen (1973). ✓

Test 3 (eikonal scaling): $\mathrm{Re}(\omega_{\mathrm{QNM}})/l \to \Omega_{\mathrm{ph}}$ (constant) for $l = 2, \ldots, 100$. ✓

For $l = 2$, the eikonal estimate gives $\mathrm{Re}(\omega) \simeq 0.703\,M^{-1}$ versus the exact Leaver value $\simeq 0.568\,M^{-1}$ [Berti et al., 2009] (24% discrepancy, expected for $l = 2$, since Eq. (18) is only asymptotically valid for $l \gg 1$).

7.3 Dispersive corrections

Table 3 gives $\Omega_{\mathrm{ph}}(\Lambda)$ and $\lambda_{\mathrm{Lyap}}(\Lambda)$ for both rings at $E = 1$.

Table 3: Eikonal QNM parameters vs Λ ($a^* = 0.9$, $M = 1$, $E = 1$), from Eq. (19).

Λ	$\Omega_{\mathrm{ph}}^{\mathrm{pro}}$	$\lambda_{\mathrm{Lyap}}^{\mathrm{pro}}$	$\Omega_{\mathrm{ph}}^{\mathrm{ret}}$	$\lambda_{\mathrm{Lyap}}^{\mathrm{ret}}$
0	0.351565	0.136080	−0.146363	0.186549
10^{-4}	0.351588	0.136106	−0.146437	0.186608
10^{-3}	0.351796	0.136341	−0.147101	0.187144
10^{-2}	0.353864	0.138663	−0.153645	0.192451

To first order (exactly proportional to E^2):

$$\begin{aligned} \text{prograde:} \quad & \frac{\delta\Omega_{\mathrm{ph}}}{\Omega_{\mathrm{ph}}^{\mathrm{GR}}} \simeq +0.657\,\Lambda E^2, \qquad \frac{\delta\lambda_{\mathrm{Lyap}}}{\lambda_{\mathrm{Lyap}}^{\mathrm{GR}}} \simeq +1.924\,\Lambda E^2, \\ \text{retrograde:} \quad & \frac{\delta|\Omega_{\mathrm{ph}}|}{|\Omega_{\mathrm{ph}}^{\mathrm{GR}}|} \simeq +5.047\,\Lambda E^2, \qquad \frac{\delta\lambda_{\mathrm{Lyap}}}{\lambda_{\mathrm{Lyap}}^{\mathrm{GR}}} \simeq +3.196\,\Lambda E^2. \end{aligned} \tag{20}$$

With $E \simeq l\,|\Omega_{\mathrm{ph}}|$, the shift of the real part of an equatorial eikonal mode grows as l^3; for the modes $m = \pm l$ the difference of the two shifts, from which the GR splitting cancels, is $D_\Lambda \simeq 0.0127\,\Lambda\,l^3$ (geometric units). For astrophysical black holes and natural values of Λ these corrections are negligible.

Remark 4. *Equation* (18) *is an eikonal ($l \gg 1$) effective approximation, not an exact result. For the dominant $l = 2$ gravitational-wave mode, a full solution of the dispersive radial wave equation is required. With the Boyer–Lindquist/ZAMO foliation used here, the horizon boundary conditions of that wave problem are not fixed by the model (the dispersive terms dominate near the horizon and remove the ingoing-wave behaviour); this is documented separately in the repository notes (A2a_SCHWARZSCHILD_BOUNDARY.md).*

8 Conclusions

We have derived four results for the quartic-dispersion Kerr model:

1. **Photon-ring shift (Result 1):** $\delta b = C(a^*, \text{ring})\,\Lambda\,E^2$, linear in Λ and quadratic in photon energy. For $a^* = 0.9$: $C_{\text{pro}} = -0.6230\,M^{-1}$, $C_{\text{ret}} = +11.495\,M^{-1}$. The ≈ 18.5-fold magnitude asymmetry reflects near-degeneracy of the ring-defining Jacobian at the retrograde ring, not the magnitude of k^4_{loc} (Remark 1).

2. **Chromatic shadow (Theorem 1):** The E^2 scaling is exact and follows from the vanishing of $g^{\mu\nu}k_\mu k_\nu$ at the GR photon ring. The frequency-dependent sensitivity scale satisfies $\Lambda_{\max} \propto \omega^{-2}$.

3. **Superradiance (Theorem 2):** The kinematic threshold $\omega = m\Omega_H$ is unchanged by Λ. The quartic term vanishes identically at the threshold because $\Omega_{\text{ZAMO}} \to \Omega_H$ at the horizon. The amplification factor away from threshold is not addressed here.

4. **Eikonal QNMs (Section 7):** We use the eikonal continuation $\omega_{\text{QNM}} \simeq \Omega_{\text{ph}}\,l - i(n+\frac{1}{2})\lambda_{\text{Lyap}}$ as an effective approximation, with first-order corrections proportional to ΛE^2 (Eq. (20)); for eikonal modes they grow as l^3. The $\Lambda \to 0$ limit is validated against exact Schwarzschild and Kerr benchmarks.

Open questions: (i) Full $l = 2$ QNM spectrum from the dispersive wave equation, which requires horizon boundary conditions that the present foliation does not fix. (ii) Dynamical consequences of Carter non-separability (Appendix A). (iii) Physical origin of Λ.

Corrections in v0.5

Version 0.5 corrects computational errors in v0.4, found while extending Section 7 to the retrograde ring. All corrected numbers are reproduced by scripts in the public repository (github.com/Kretski/Lambda-model, `notes/`: `eikonal_d_lambda.py`, `paper2_table3_corrected.py`; notes `EIKONAL_D_LAMBDA.md`), using two independent methods (numerical root finding and implicit differentiation with exact symbolic derivatives), which agree to ten digits.

1. **Table 1, b columns.** The values printed in v0.4 did not come from the v0.4 code and, for the retrograde ring, moved opposite to the paper's own coefficient C_{ret}. They are replaced by the output of the v0.4 code. The radii and the coefficients C_{pro}, C_{ret} (Result 1) are unchanged and confirmed.

2. **Table 3 and the Ω_{ph} correction.** In v0.4, $\dot\phi$ and $\dot t$ were computed from the general-relativistic part of H only, at the Λ-shifted ring. This reproduces the v0.4 values exactly (e.g. 0.350800 at $\Lambda = 10^{-2}$ and $-0.219\,\Lambda$) but is inconsistent with the dispersive Hamiltonian. With all terms of H the correction is $+0.657\,\Lambda E^2$ (prograde).

3. **Lyapunov exponent.** The v0.4 expression was mis-normalised (it gave 0.2357 instead of $1/(3\sqrt{3}) = 0.1925$ for Schwarzschild in the code used for Table 3). It is replaced by Eq. (19), which reproduces the Schwarzschild value exactly. The coefficient $+0.066\,\Lambda$ of v0.4 becomes $+1.924\,\Lambda E^2$.

4. **Energy dependence.** v0.4 evaluated the eikonal corrections at $E = 1$. Since they scale as E^2 and $E \simeq l\,\Omega_{\rm ph}$ for an eikonal mode, they grow as l^3; Eqs. (18)–(20) now state this.

5. **Remarks 1 and in Section 7.** The v0.4 prediction $\lambda^{\rm ret}_{\rm Lyap}/\lambda^{\rm pro}_{\rm Lyap} \approx 0.12$ is not borne out; the ratio is 1.37 at $\Lambda = 0$.

Results (1)–(3) of the abstract (photon-ring coefficients, chromatic E^2 scaling, superradiant threshold) are unaffected.

A Carter Non-Separability: Exact Symbolic Result

For completeness we record the exact separability test, which motivates item (ii) of the open questions in Section 7.

The standard Kerr Hamiltonian ($\Lambda = 0$) is fully separable via the Carter constant $Q_{\rm Carter}$:

$$\{Q_{\rm Carter},\, H_{\rm GR}\} = 0 \quad (\Lambda = 0). \tag{21}$$

This was verified to exact rational arithmetic using SymPy.

For $\Lambda > 0$ and $a \neq 0$, exact symbolic computation gives:

$$\{Q_{\rm Carter},\, H_\Lambda\} = -\frac{624358\ldots}{2809107\ldots} \neq 0, \tag{22}$$

an exact rational number, confirming that the classical Carter invariant is not conserved by the dispersive Hamiltonian.

Interpretation. Non-vanishing of the Poisson bracket establishes that $Q_{\rm Carter}$ is not a constant of motion for H_Λ. It does *not* imply chaos: a new Λ-dependent invariant $Q_{\rm eff}(\Lambda)$ may exist near $Q_{\rm Carter}$.

Dynamical tests. Poincaré sections (197 crossings per Λ value, full 4D Hamiltonian integration with corrected Kerr metric) show closed C-shaped curves at all tested $\Lambda \in [0, 10^{-4}]$. Lyapunov exponents converge to $\lambda_{\rm max} \to 0$ for $\Lambda \leq 10^{-3}\,M^2$, consistent with quasi-integrable or regular dynamics at small Λ. Whether chaos appears at larger Λ requires extended integration times and is deferred to Paper 3.

Superradiance connection. The Carter non-separability is *not* required for the superradiance result in Section 6: that proof depends only on the horizon geometry ($\Omega_{\rm ZAMO} \to \Omega_H$), which holds independently of separability.

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